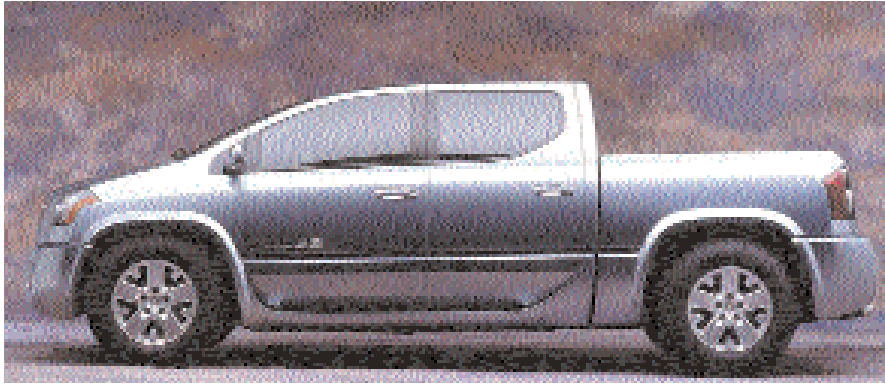




Take a ride in the MAXXcab and headbang to metal, while your parents listen to oldies

to life and enormously entertaining.

Holosonic Labs is working on another interesting application at the Boston Museum of Science that allows the intended listeners to understand and hear explanations, without raising the ambient sound levels. The idea is that museum exhibits can be discreetly wired up with tiny speaker domes that can unobtrusively, provide explanations. There are also other interesting applications that they are looking

at, such as private messaging using this system without headphones, special effects at presentations as well as special sound theme parks that could put up animated sound displays similar to today's light shows. Holosonic has installed their Audio Spotlight system at Tokyo's Sega Joyopolis theme park. One of the most interesting applications though, is the one fitted in the DaimlerChrysler MAXXcab prototype truck. Holosonic put in four individual Audio Spotlights into the truck to let all the passengers enjoy their own choice of music. This has been successful enough for them to start looking at marketing this commercially.

American Technology Corporation has also been looking at emerging consumer and non-consumer markets, targeting consumer electronics and home entertainment as well as a decidedly niche usage in the military. The US Navy has installed sound beaming technology on the deck of an Aegis-class Navy destroyer, and is looking at this as a substitute to the radio operator's headphones. To keep track of what's going on around

Beaming alternatively

There are other alternative approaches to creating targeted audio, other than the ultrasound modulation technique. One is the parabolic dish approach that essentially uses antennae to focus and direct sound. Here, a relatively omni-directional loudspeaker is placed at the focal point of a parabolic dish pointing towards it. When the loudspeaker generates the sound signal, it acts as a point source, emitting waves that reflect off the parabolic dish which is pointed towards a particular direction. This is very much in use, but the problem here is that the size of the parabolic dish required to accommodate the longer wavelengths of lower frequencies is too large.

them, as well as the radio chatter, users generally keep one ear off the radio headset. With this system, all the information can be piped into the operator's ears, without them having to wear bulky headsets.

There are even bigger applications as far as the military goes. One of them is a very loud hailing device to catch the attention of ships further away, though it does not use ultrasound as the underlying technology. Still more interesting is the idea of non-lethal acoustic assault rifles that fire sound pulses. Pump up the normal sound being transmitted to about 150 decibels or greater, and you could fire out pulses that could disorient human targets, even causing them severe physical pain. The weapon could also be fine-tuned to bring on further discomfort. American Technology Corp is looking at Jeep mounted units that could be deployed as a mob deterrent.

Sounding right

There are even more interesting applications in the pipeline—car-based safety audio systems, discreet speaker phones and many more such interesting applications. Over the next few years, the way we experience sound may change dramatically. We will reproduce sound just the way we experience it in the real world. Is it time to trash your boom box? Well, not yet, but pretty soon, the next big bang in sound is going to come straight out of the air, riding on inaudible ultrasound. ■

SRINIVASAN RAMAKRISHNAN
(srinivasan_ramakrishnan@thinkdigit.com)